

Probability Theory Part II

Variables:

Let's say we want to define a variable of height of a person.

It has a fixed value. It is deterministic.

Let's say we want to define a variable for the current weather temperature.

It has a fixed value. It is deterministic.

Let's say we want to define a variable for the outcome of rolling a dice.

It is not fixed and may have any value from 1 to 6. It is Probabilistic.

An Ordinary Variable or Deterministic Variable represents fixed information.

A Random Variable represents uncertain outcomes in the future.

Examples:

- If the outcome is fixed and known beforehand is ordinary Variable
- If the outcome is unknown and depends on chance is Random Variable
- The speed of your car at a given time is Ordinary Variable
- The speed of the next car passing by on the highway is Random Variable

Definition

A **random variable** is a quantity whose value is determined by the outcomes of the random experiments.

We can define a random variables as a function mapping each outcome in a probability space to a real number.

Given a probability space $(\Omega, \mathcal{F}, P)$, a **random variable** X is a function from the sample space Ω to the real numbers $\mathbb{R}$.

Once the outcome $\omega \in \Omega$ of the experiment is revealed, the corresponding $X(\omega)$ is known as a realization of the random variable.

Definition

A **random variable** is also called a **chance variable**, a **stochastic variable** or a **variate**.

Random variables are usually denoted by capital letters such as X , Y , Z .

Types of Random Variables:

There are two types of variables:

1. Discrete Random Variable
2. Continuous Random Variable

Types of Random Variables:

There are two types of variables:

1. Discrete Random Variable

- a) Number of defective bulbs in a box of 100 bulbs,
- b) Number of calls received at a helpline center in one hour,
- c) Number of cars crossing a toll plaza in 10 minutes

2. Continuous Random Variable

- a) Example: Daily Temperature in a City,
- b) Time taken by a student to solve a math problem,
- c) Amount of rainfall in a day

Example:

Suppose that our experiment consists of tossing 3 fair coins.

If we let Y denote the number of heads that appear,

then Y is a random variable taking on one of the values 0, 1, 2 and 3 then,

$$Y(T, T, T) = 0, Y(H, T, T) = 1$$

The outcomes related to $Y = 1$ are $(H, T, T), (T, H, T), (T, T, H)$.

Example:

The sample space giving a detailed description of each possible outcome when three electronic components are tested may be written

$$S = \{NNN, NND, NDN, DNN, NDD, DND, DDN, DDD\},$$

where N denotes non-defective and D denotes defective.

One is naturally concerned with the number of defectives that occur. Thus, each point in the sample space will be assigned a numerical value of 0, 1, 2, or 3.

These values are random quantities determined by the outcome of the experiment.

Hence, the random variable X represents the number of defective items when three electronic components are tested.

Notations:

We shall use a capital letter, say X , to denote a random variable and its corresponding small letter, x in this case, for one of its values.

In the electronic component testing, we notice that the random variable X assumes the value 2 for all elements in the subset

$$E = \{DDN, DND, NDD\}$$

of the sample space S .

That is, each possible value of X represents an event that is a subset of the sample space for the given experiment.

Example:

Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2 and 3 with probability

$$P(Y = 0) = P\{(T, T, T)\} = \frac{1}{8}$$

$$P(Y = 1) = P\{(H, T, T), (T, H, T), (T, T, H)\} = \frac{3}{8}$$

$$P(Y = 2) = P\{(H, H, T), (H, T, H), (T, H, H)\} = \frac{3}{8}$$

$$P(Y = 3) = P\{(H, H, H)\} = \frac{1}{8}$$

Since Y must take on one of the values 0 through 3, we must have

$$1 = P\left(\bigcup_{i=0}^3 [Y = i]\right) = \sum_{i=0}^3 P(Y = i)$$

which, of course, is in accord with the preceding probabilities.

Discrete Probability Distribution:

The set of ordered pairs $(x, f(x))$ is a **probability function**, **probability mass function**, or **discrete probability distribution** of the discrete random variable X if, for each possible outcome x ,

1. $f(x) = P(X = x) = P\{\zeta \in \Omega | X(\zeta) = x\}$.
2. $f(x) \geq 0$,
3. $\sum_x f(x) = 1$,

Example:

A shipment of 20 similar laptop computers to a retail outlet contains 3 that are defective. If a school makes a random purchase of 2 of these computers, find the probability distribution for the number of defectives.

Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. Then x can only take the numbers 0, 1, and 2.

Now

$$f(0) = P(X = 0) = \frac{\binom{3}{0} \binom{17}{2}}{\binom{20}{2}} = \frac{68}{95},$$

$$f(1) = P(X = 1) = \frac{\binom{3}{1} \binom{17}{1}}{\binom{20}{2}} = \frac{51}{190}$$

$$f(2) = P(X = 2) = \frac{\binom{3}{2} \binom{17}{0}}{\binom{20}{2}} = \frac{3}{190}$$

Thus, the probability distribution of X is

x		0	1	2
$f(x)$		$\frac{68}{95}$	$\frac{51}{190}$	$\frac{3}{190}$

Example (revisited):

Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2 and 3 with probability

$$P(Y = 0) = P\{(T, T, T)\} = \frac{1}{8}$$

$$P(Y = 1) = P\{(H, T, T), (T, H, T), (T, T, H)\} = \frac{3}{8}$$

$$P(Y = 2) = P\{(H, H, T), (H, T, H), (T, H, H)\} = \frac{3}{8}$$

$$P(Y = 3) = P\{(H, H, H)\} = \frac{1}{8}$$

Find $P(Y \geq 2)$, $P(Y < 2)$, $P(2 \leq Y \leq 3)$.

Two Requirements for a Probability Distribution Function:

1. The sum of the probabilities of all the events in the sample space must equal 1; that is, $\sum f(x) = 1$.
2. The probability of each event in the sample space must be between or equal to 0 and 1. That is, $0 \leq f(x) \leq 1$.

Example:

Determine whether each distribution is a probability distribution.

<i>a.</i>	X	4	6	8	10
	$P(X)$	-0.6	0.2	0.7	1.5

<i>b.</i>	X	1	2	3	4
	$P(X)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

<i>c.</i>	X	8	9	12
	$P(X)$	$\frac{2}{3}$	$\frac{1}{6}$	$\frac{1}{6}$

<i>d.</i>	X	1	3	5	7	9
	$P(X)$	0.3	0.1	0.2	0.4	-0.7

Bernoulli random variables

Bernoulli random variables are used to model experiments that have two possible outcomes.

By convention we usually represent an outcome by 0 and the other outcome by 1.

A canonical example is flipping a biased coin, such that the probability of obtaining heads is p .

If we encode heads as 1 and tails as 0, then the result of the coin flip corresponds to a Bernoulli random variable with parameter p .

Bernoulli random variables

Definition:

The pmf (probability mass function) of a Bernoulli random variable with parameter $p \in [0, 1]$ is given by

$$\begin{aligned}P(X = 0) &= 1 - p, \\P(X = 1) &= p.\end{aligned}$$

Example:

Imagine that we take a biased coin and flip it until we obtain heads. If the probability of obtaining heads is p and the flips are independent, then the probability of having to flip k times is

$$\begin{aligned} P(k \text{ flips}) &= P(1st = tails, \dots, k - 1th = tails, kth = heads). \\ &= P(1st = tails) \dots P(k - 1th = tails) P(kth = heads) \\ &= (1 - p)^{k-1} p. \end{aligned}$$

This reasoning can be applied to any situation in which a random experiment with a fixed probability p is repeated until a particular outcome occurs, as long as the independence assumption is met.

In such cases the number of repetitions is modeled as a geometric random variable.

Definition:

The probability mass function of a **geometric random variable** with parameter p is given by

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

Definition:

In a binomial experiment, the probability of exactly X successes in n trials is

$$P(X = x) = \frac{n!}{x! (n - x)!} p^x (1 - p)^{n-x}$$

Example:

If we flip a biased coin n times, what is the probability that we obtain exactly k heads if the flips are independent, and the probability of heads is p

Cumulative Distribution Function

The **cumulative distribution function** $F(x)$ of a discrete random variable X with probability distribution $f(x)$ is

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

Example:

If a car agency sells 50% of its inventory of a certain foreign car equipped with side airbags, find a formula for the probability distribution of the number of cars with side airbags among the next 4 cars sold by the agency.

Let X define a random variable representing the number of cars with side airbags

$$f(x) = P(X = x) = \frac{1}{16} \binom{4}{x}, x = 0, 1, 2, 3, 4$$

Example:

Find the cumulative distribution function of the random variable X in the previous example. Using $F(x)$, verify that $f(2) = 3/8$.

Solution : Direct calculations of the probability distribution give

$$f(0) = \frac{1}{16}, f(1) = \frac{1}{4}, f(2) = \frac{3}{8}, f(3) = \frac{1}{4}, \text{ and } f(4) = \frac{1}{16}.$$

Therefore,

- $F(0) = f(0) = \frac{1}{16}$
- $F(1) = f(0) + f(1) = \frac{5}{16}$
- $F(2) = f(0) + f(1) + f(2) = \frac{11}{16}$
- $F(3) = f(0) + f(1) + f(2) + f(3) = \frac{15}{16}$
- $F(4) = f(0) + f(1) + f(2) + f(3) + f(4) = 1.$

Example:

Find the distribution function of the sum of outcomes of two dice rolled once.

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x_i	$f(x_i)$	$F(x_i)$
2	1/36	1/36
3	2/36	3/36
4	3/36	6/36
5	4/36	10/36
6	5/36	15/36
7	6/36	21/36
8	5/36	26/36
9	4/36	30/36
10	3/36	33/36
11	2/36	35/36
12	1/36	36/36

Example:

Find the distribution function of the sum of the outcomes of two dice rolled once.

Use the probability distribution to find the probabilities of obtaining

1. A sum of 8 and 11
2. A sum that is greater than 8
3. A sum that is greater than 5 but less than or equal to 10

x_i	$f(x_i)$	$F(x_i)$
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3	2/36	3/36
4	3/36	6/36
5	4/36	10/36
6	5/36	15/36
7	6/36	21/36
8	5/36	26/36
9	4/36	30/36
10	3/36	33/36
11	2/36	35/36
12	1/36	36/36

Properties of Cumulative Distribution

1. $\lim_{x \rightarrow -\infty} F(x) = 0$
2. $\lim_{x \rightarrow \infty} F(x) = 1$
3. If $a < b$ then $F(a) \leq F(b)$, that is it is non-decreasing.
4. $F(x)$ is right-hand continuous, that is $\lim_{h \rightarrow 0^+} F(x + h) = F(x)$
5. $F(x) = \sum_i f(x_i)$
6. $F(b) - F(a) = P(a < x \leq b)$

Continuous Random Variable

A random variable X is defined to be **continuous** if it can assume every possible value in an interval $[a, b]$.

Example:

Height of a person

Temperature at a place

Amount of rainfall

Continuous Random Variable

- We will be computing probabilities for various intervals of continuous random variables such as $P(a < X < b)$, $P(W \geq c)$, and so forth.
- Note that when X is continuous,
$$P(a < X \leq b) = P(a < X < b) + P(X = b) = P(a < X < b).$$
- That is, it does not matter whether we include an endpoint of the interval or not.
- Although the probability distribution of a continuous random variable cannot be presented in tabular form, it can be stated as a formula.
- Such a formula is a function of the numerical values of the continuous random variable X and as such will be represented by the functional notation $f(x)$.
- $f(x)$ is usually called the **probability density function**, or simply the **density function**, of X .

Probability Density Function:

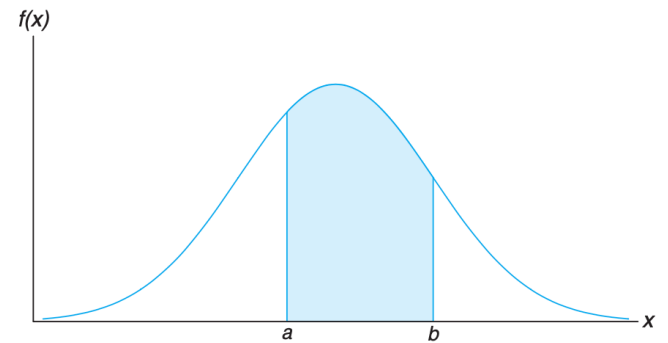
The function $f(x)$ is called the **probability density function**, abbreviated to p.d.f., of the continuous random variable X , if it satisfies the following properties:

1. $f(x) \geq 0, \forall x$

2. $\int_{-\infty}^{\infty} f(x)dx = 1$

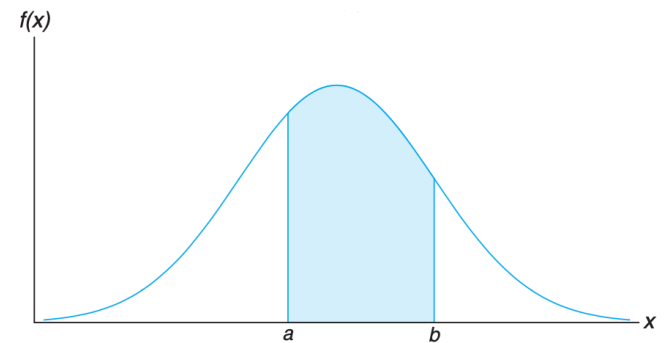
3. $P(a < x \leq b) = \int_{-\infty}^b f(x)dx - \int_{-\infty}^a f(x)dx$

$$= \int_a^b f(x)dx$$



Probability Density Function:

The key idea behind a **Probability Density Function (PDF)** is that it describes the *relative likelihood* of a continuous random variable taking on a particular value. But for a continuous variable, the probability at a single point is always **zero**. Instead, the PDF gives a measure of how densely probability is packed around that point.



Example:

Suppose that the error in the reaction temperature, in °C, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

- (a) Verify that $f(x)$ is a density function.
- (b) Find $P(0 < X \leq 1)$.

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(a) Verify that $f(x)$ is a density function.

Obviously, $f(x) \geq 0$. To verify the next condition, we have

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-1}^2 \frac{x^2}{3} dx = \frac{x^3}{9} \Big|_{-1}^2 = 1$$

Example:

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(b) Find $P(0 < X \leq 1)$.

$$P(0 < X \leq 1) = \int_0^1 f(x) dx = \left. \frac{x^3}{9} \right|_0^1 = \frac{1}{9}$$

Example:

Let X be a random variable having probability density function

$$f(x) = \begin{cases} cx, & 0 \leq x \leq 2, \\ 0, & \text{elsewhere} \end{cases}$$

Find

- a) The value of c
- b) $P(X < 0.5)$
- c) $P(X > 1)$
- d) $P\left(\frac{1}{2} < X < \frac{3}{2}\right)$
- e) $P(X = 1)$

Cumulative Distribution Function

The **cumulative distribution function** $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt, \quad \text{for } -\infty < x < \infty$$

Example:

Let $f(x)$ be a density function defined as follows:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

Compute the cumulative distribution function $F(x)$ and use it to evaluate $P(0 < X \leq 1)$.

Example:

Let $f(x)$ be a density function defined as follows:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

Compute the cumulative distribution function $F(x)$ and use it to evaluate $P(0 < X \leq 1)$.

We have to evaluate $F(x)$ for the intervals, $x \leq -1$, $-1 < x < 2$ and $x \geq 2$.

$$\text{For } x \leq -1: F(x) = \int_{-\infty}^{-1} 0 \, dx = 0$$

$$\text{For } -1 < x < 2: F(x) = \int_{-\infty}^{-1} 0 \, dx + \int_{-1}^x \frac{x^2}{3} \, dx = 0 + \frac{x^3}{9} \Big|_{-1}^x = \frac{x^3}{9} - \left(-\frac{1}{9}\right) = \frac{x^3+1}{9}$$

$$\text{For } x \geq 2: F(x) = \int_{-\infty}^{-1} 0 \, dx + \int_{-1}^2 \frac{x^2}{3} \, dx + \int_2^x 0 \, dx = 1$$

Example:

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$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

Compute the cumulative distribution function $F(x)$ and use it to evaluate $P(0 < X \leq 1)$.

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$$\text{For } x \geq 2: F(x) = \int_{-\infty}^{-1} 0 \, dx + \int_{-1}^2 \frac{x^2}{3} \, dx + \int_2^x 0 \, dx = 1$$

Therefore,

$$F(x) = \begin{cases} 0, & x \leq -1 \\ \frac{x^3+1}{9}, & -1 < x < 2 \\ 1, & x \geq 2 \end{cases}$$

Example:

Let $f(x)$ be a density function defined as follows:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere} \end{cases}$$

Compute the cumulative distribution function $F(x)$ and use it to evaluate $P(0 < X \leq 1)$.

Therefore,

$$F(x) = \begin{cases} 0, & x \leq -1 \\ \frac{x^3 + 1}{9}, & -1 < x < 2 \\ 1, & x \geq 2 \end{cases}$$

Now,

$$P(0 < X \leq 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}.$$

Note:

$$1. \frac{d}{dx}F(x) = f(x)$$

$$2. P(a < X \leq b) = F(b) - F(a)$$

Example:

Let X be a random variable having probability distribution function

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{elsewhere} \end{cases}$$

Find $P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$

Solution:

$$\begin{aligned} P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right) &= \frac{P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)} = \frac{\int_{1/3}^{1/2} 2x \, dx}{\int_{1/3}^{2/3} 2x \, dx} \\ &= \frac{x^2 \Big|_{1/3}^{1/2}}{x^2 \Big|_{1/3}^{2/3}} = \frac{\frac{1}{4} - \frac{1}{9}}{\frac{4}{9} - \frac{1}{9}} = \frac{\frac{5}{36}}{\frac{3}{9}} = \frac{5}{12} \end{aligned}$$

Example:

A continuous random variable has the cumulative distribution function as follows

$$F(X) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{2x^2}{5}, & \text{for } 0 < x \leq 1, \\ -\frac{3}{5} + \frac{2}{5}\left(3x - \frac{x^2}{2}\right), & \text{for } 1 < x \leq 2, \\ 1, & \text{for } x > 2. \end{cases}$$

Find the probability density function and evaluate $P(|X| < 1.5)$

Example:

Answer:

$$f(x) = \begin{cases} 0, & \text{for } x < 0, \\ \frac{4x}{5}, & \text{for } 0 < x \leq 1, \\ \frac{2}{5}(3 - x), & \text{for } 1 < x \leq 2, \\ 0, & \text{for } x > 2. \end{cases}$$

$$P(|X| < 1.5) = \frac{3}{4}$$

Expectation and Variance of a Random Variable

Example:

Suppose that our experiment consists of tossing 3 fair coins. If we let Y denote the number of heads that appear, then Y is a random variable taking on one of the values 0, 1, 2 and 3 with probability

Y	0	1	2	3
Frequency	1	3	3	1
Probability mass function	1/8	3/8	3/8	1/8

Find the mean of Y .

Expectation

One of the most important concepts in probability theory is that of the expectation of a random variable.

If X is a discrete random variable having a probability mass function $f(x)$, then the **expectation**, or the **expected value**, or the **mean**, of X , denoted by $E[X]$, is defined by

$$\mu = E(X) = \sum_x x f(x)$$

In words, the expected value of X is a weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it.

Expectation

If X is a continuous random variable having a probability density function $f(x)$, then the **expectation**, or the **expected value**, or the **mean**, of X , denoted by $E[X]$, is defined by

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

Example:

Assume that 1 fair coin was tossed twice, we find that the sample space for our experiment is

$$S = \{HH, HT, TH, TT\}.$$

Since the 4 sample points are all equally likely, for number of heads per toss it follows that

$$P(X = 0) = P(TT) = 1/4, \quad P(X = 1) = P(TH) + P(HT) = 1/2, \text{ and} \\ P(X = 2) = P(HH) = 1/4,$$

Now, these probabilities are just the relative frequencies for the given events in the long run. Therefore,

$$\mu = E(X) = (0) \left(\frac{1}{4}\right) + (1) \left(\frac{1}{2}\right) + (2) \left(\frac{1}{4}\right) = 1.$$

This result means that a person who tosses 2 coins over and over again will, on the average, get 1 head per toss.

Example:

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

Example:

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

Solution: Let X represent the number of good components in the sample. The probability distribution of X is

$$f(x) = \frac{\binom{4}{x} \binom{3}{3-x}}{\binom{7}{3}}, x = 0, 1, 2, 3$$

Simple calculations yield $f(0) = 1/35$, $f(1) = 12/35$, $f(2) = 18/35$, and $f(3) = 4/35$. Therefore,

$$\mu = E(X) = (0) \left(\frac{1}{35} \right) + (1) \left(\frac{12}{35} \right) + (2) \left(\frac{18}{35} \right) + (3) \left(\frac{4}{35} \right) = \frac{12}{7} = 1.7.$$

Thus, if a sample of size 3 is selected at random over and over again from a lot of 4 good components and 3 defective components, it will contain, on average, 1.7 good components.

Example:

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere} \end{cases}$$

Find the expected life of this type of device.

$$\begin{aligned} \mu = E(X) &= \int_{100}^{\infty} \frac{20,000}{x^2} dx \\ &= 200 \end{aligned}$$

Example:

Let X be a random variable with probability density function

$$f(x) = \begin{cases} 6(x - x^2), & 0 \leq x \leq 1, \\ 0, & \text{elsewhere,} \end{cases}$$

Find the expectation.

Solution

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_{-\infty}^0 0 \, dx + \int_0^1 6(x^2 - x^3) \, dx \\ &= 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \left[\frac{1}{3} - \frac{1}{4} \right] = 6 \left(\frac{1}{12} \right) = \frac{1}{2} \end{aligned}$$

Properties of Expected value:

If X is a discrete random variable, then the expected value of X holds the following properties:

1. $E(a) = a$
2. $E(aX) = aE(X)$
3. $E(aX + b) = aE(X) + b$
4. $E(X - \mu) = 0$ (*prov it!*)

Expectation of a function of a random variable

Let $g(X)$ be a function of a random variable X . Then $g(X)$ is also a random variable and also has an expected value:

$$\begin{aligned} E(g(X)) &= g(x_1)f(x_1) + \cdots + g(x_n)f(x_n) \\ &= \sum_i g(x_i)f(x_i) \end{aligned}$$

provided that the series converges, where $f(x)$ be probability distribution function of X .

Similarly, if X is a continuous random variable with probability density function $f(x)$, then

$$E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Provided the integral exists.

Example 4.4: Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

Solution: By Theorem 4.1, the attendant can expect to receive

$$\begin{aligned}
 E[g(X)] &= E(2X - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\
 &= (7) \left(\frac{1}{12} \right) + (9) \left(\frac{1}{12} \right) + (11) \left(\frac{1}{4} \right) + (13) \left(\frac{1}{4} \right) \\
 &\quad + (15) \left(\frac{1}{6} \right) + (17) \left(\frac{1}{6} \right) = \$12.67.
 \end{aligned}$$



Example 4.5: Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = 4X + 3$.

Solution: By Theorem 4.1, we have

$$E(4X + 3) = \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx = 8.$$



Example:

Let $g(X) = (X - \mu)^2$, where μ is the population mean, then

$$\begin{aligned} E(g(X)) &= \sum (x_i - \mu)^2 f(x) \\ &= \text{Var}(X) \\ &= E(X^2) - (E(X))^2 \end{aligned}$$

Variance and Standard Deviation

Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$
$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

The positive square root of the variance, σ , is called the **standard deviation** of X .

Example:

Let X be a random variable with probability density function

$$f(x) = \begin{cases} 6(x - x^2), & 0 \leq x \leq 1, \\ 0, & \text{elsewhere,} \end{cases}$$

Find the variance and standard deviation.

Solution: We already calculated $E(X) = 1/2$,

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx \\ &= \int_{-\infty}^0 0 dx + \int_0^1 6(x^3 - x^4) dx \\ &= 6 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = 6 \left[\frac{1}{4} - \frac{1}{5} \right] = 6 \left(\frac{1}{20} \right) = \frac{3}{10} \\ \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \frac{3}{10} - \frac{1}{4} = \frac{6 - 5}{20} = \frac{1}{20} \end{aligned}$$

Properties of Variance

1. $Var(X) \geq 0$
2. $Var(a) = 0$
3. $Var(aX) = a^2 Var(X)$
4. $Var(aX + b) = a^2 Var(X)$

Normal Distribution

The most important continuous probability distribution in the entire field of statistics is the **normal distribution**. Its graph, called the **normal curve**, is the **bell-shaped curve**.

The normal distribution is often referred to as the **Gaussian distribution**.

Normal Distribution

The mathematical equation for the probability distribution of the normal variable depends on the two parameters μ and σ , its mean and standard deviation, respectively. Hence, we denote the values of the density of X by $n(x; \mu, \sigma)$ or $N(\mu, \sigma^2)$.

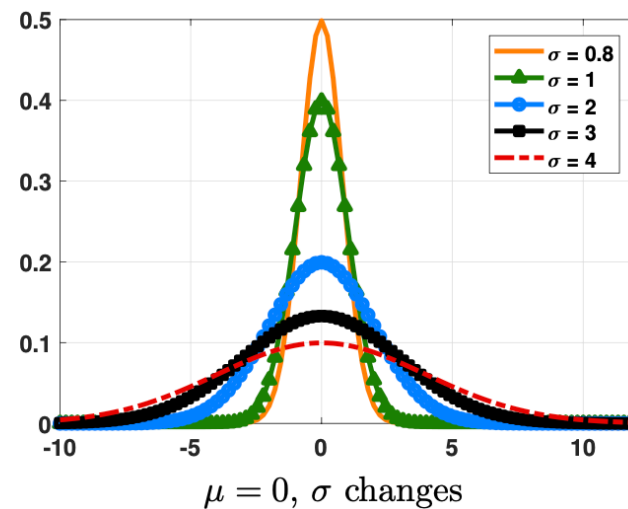
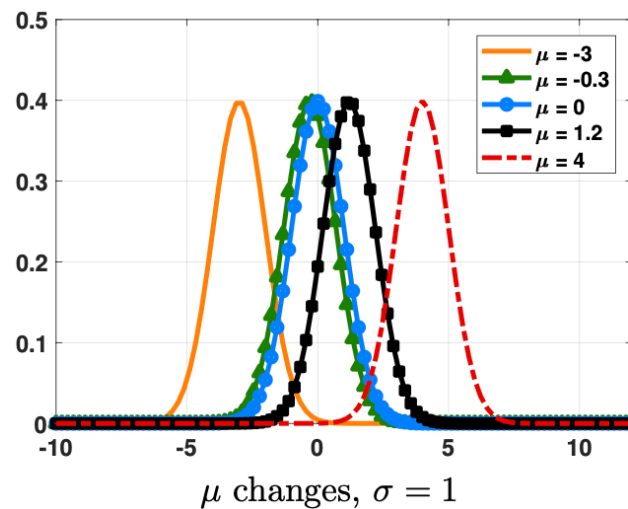
The density of the normal random variable X , with mean μ and variance σ^2 , is

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, -\infty < x < \infty,$$

Normal Distribution

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Properties of Normal Distribution

The following properties of the normal curve:

- The mode, which is the point on the horizontal axis where the curve is a maximum, occurs at $x = \mu$.
- The curve is symmetric about a vertical axis through the mean μ .
- The curve has its points of inflection at $x = \mu \pm \sigma$; it is concave downward if $\mu - \sigma < X < \mu + \sigma$ and is concave upward otherwise.
- The normal curve approaches the horizontal axis asymptotically as we proceed in either direction away from the mean.
- The total area under the curve and above the horizontal axis is equal to 1.
- The odd order moments about mean are all zero, whereas, the even order moments about mean are $\mu_{2n} = (2n - 1)(2n - 3) \dots 5.3.1\sigma^{2n}$.
- If X is $N(\mu, \sigma^2)$ and if $Y = a + bX$, then Y is $N(a + b\mu, b^2\sigma^2)$

Properties of Normal Distribution

The following properties of the normal curve:

- The odd-order moments about the mean are all zero.
- If X is $N(\mu, \sigma^2)$ and if $Y = a + bX$, then Y is $N(a + b\mu, b^2\sigma^2)$, the pdf of Y is

$$h(Y) = \frac{1}{|b|\sigma\sqrt{2\pi}} e^{-[y-(a+b\mu)]^2/2b^2\sigma^2}$$

by using the formula $h(y) = f(x) \left| \frac{dx}{dy} \right|$.

This implies that if $Z = \frac{X-\mu}{\sigma}$, then Z is $N(0,1)$.

- The sum of independent normal variables is a normal variable. That is, if X_1 is $N(\mu_1, \sigma_1)$ and X_2 is $N(\mu_2, \sigma_2)$, then for independent X_1 and X_2 , $X_1 + X_2$ is $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Properties of Normal Distribution

The following properties of the normal curve:

- Empirical Rule: No matter what values of μ and σ are area under the normal curve remain in certain fixed proportion within a specified number of standard deviations on either sides of μ
 - $\mu \pm \sigma$ will always contain 68.26%
 - $\mu \pm 2\sigma$ will always contain 95.44%
 - $\mu \pm 3\sigma$ will always contain 99.73%
- The moment generating function about origin is $M_o(t) = e^{\mu t + \frac{1}{2}t^2\sigma^2}$

Normal Distribution

The mode, mean and median of $n(x; \mu, \sigma)$ is μ and the variance is σ^2 . Hence, the standard deviation is σ .

Normal Distribution

The distribution of a normal random variable with mean 0 and variance 1 is called a **standard normal distribution**.

Standard Normal (Standard Gaussian)

The standard Gaussian (or standard normal) random variable Z has a PDF

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

That is, $Z \sim N(0,1)$ is a Gaussian with $\mu = 0$ and $\sigma^2 = 1$.

The distribution function will be

$$\Phi(z) = P(Z < z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

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The distribution function will be

$$\Phi(z) = P(Z < z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$
$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1$$

Normal Distribution

To calculate the probability over the interval $[a, b]$ is

$$P(a \leq X \leq b) = \int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx .$$

But integrals of this type cannot be solved by ordinary means. The numerical approximations of the standard normal distribution of some areas have been calculated and compiled in form of a table.

Normal Distribution

Fortunately, we can transform all the observations of any normal random variable X into the standard normal random variable Z with mean 0 and variance 1.

This can be done using the transformation

$$Z = \frac{X - \mu}{\sigma}$$

Whenever X assumes a value x , the corresponding value of Z is given by
 $z = (x - \mu)/\sigma$.

Therefore, if X falls between the values $x = x_1$ and $x = x_2$, the random variable Z will fall between the corresponding values $z_1 = (x_1 - \mu)/\sigma$ and $z_2 = (x_2 - \mu)/\sigma$.

That is,

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \int_{z_1}^{z_2} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = P(z_1 \leq Z \leq z_2)$$

Standard Normal (Standard Gaussian)

Every normally distributed random variable X with mean $= \mu$ and variance $= \sigma^2$ can be transformed into a new random variable Z using the expression

$$Z = \frac{X - \mu}{\sigma}$$

Then,

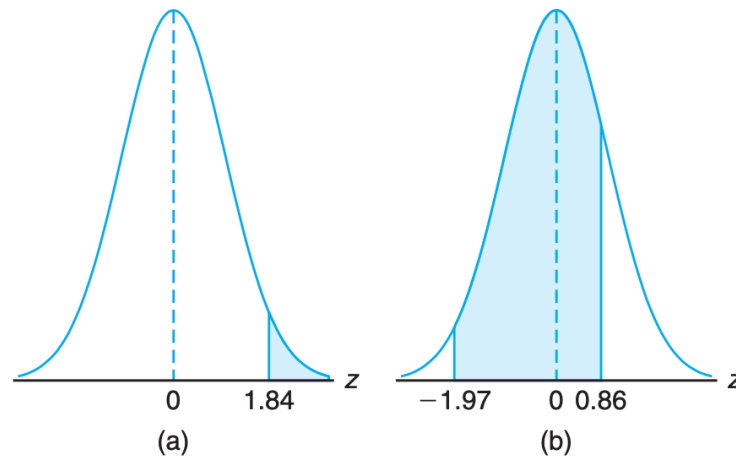
- $P(X < x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$
- $P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$
- $\Phi(-z) = 1 - \Phi(z)$

Example:

Given a standard normal distribution, find the area under the curve that lies

(a) to the right of $z = 1.84$ and

(b) between $z = -1.97$ and $z = 0.86$.



Example:

If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$, find

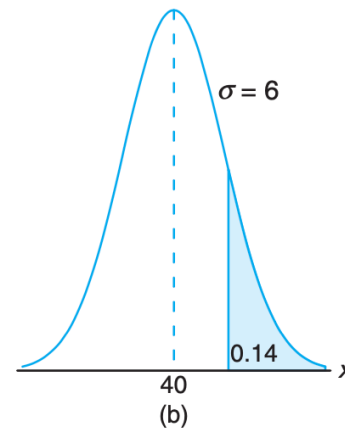
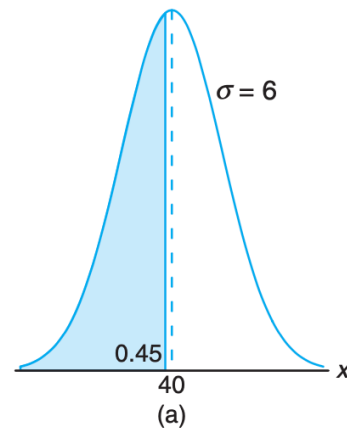
a) $P\{2 < X < 5\}$; [ans:0.3747]

b) $P\{X > 0\}$;

Example:

Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has

- (a) 45% of the area to the left and
- (b) 14% of the area to the right.



Example:

A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

Example:

In an industrial process, the diameter of a ball bearing is an important measurement. The buyer sets specifications for the diameter to be 3.0 ± 0.01 cm. The implication is that no part falling outside these specifications will be accepted. It is known that in the process the diameter of a ball bearing has a normal distribution with mean $\mu = 3.0$ and standard deviation $\sigma = 0.005$. On average, how many manufactured ball bearings will be scrapped?

Normal Approximation to Binomial Distribution

If X is a binomial random variable with mean $\mu = np$ and variance $\sigma^2 = npq$, then the limiting form of the distribution of

$$Z = \frac{X - np}{\sqrt{npq}},$$

As $n \rightarrow \infty$, is the standard normal distribution $n(z; 0, 1)$.

Or in other words, we can get a close approximation by normal distribution when n is sufficiently large and both p and q are not close to zero.

As a rule of thumb, good approximation is obtained if both np and nq are greater or equal to 5.

Example:

A fair coin is tossed 20 times. Find the probability that the number of heads occurring is between 10 and 14 inclusive by approximating using normal distribution.

Solution:

$$p = 0.5, q = 0.5, n = 20$$
$$\Rightarrow \mu = np = 10, nq = 10, \sigma^2 = npq = 5$$

We want to find the $P(10 \leq X \leq 14)$. Since we are approximating it with a continuous random variable, we will be replacing the interval by $9.5 < X < 14.5$.

$$\text{Now, } z_1 = \frac{9.5-10}{2.24} = -0.22, z_2 = 2.01. \text{ So,}$$
$$P(10 \leq X \leq 14) = P(-0.22 \leq Z \leq 2.01)$$
$$= 0.9778 - 0.4129 = 0.5649$$

The Central Limit Theorem

The Central Limit Theorem says that if you take a large enough sample from any population (with a finite mean and variance), the sampling distribution of the sample mean will be approximately Normal, no matter what the original population distribution looks like.

In other words, means are normally distributed.

The Central Limit Theorem

In mathematical terms:

If: $X_1, X_2, \dots, X_n$ are independent and identically distributed random variables

Each has mean μ and variance σ^2 .

Let $\bar{X} = \frac{1}{n} \sum X_i$ be the sample mean

Then:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow N(0,1) \text{ as } n \rightarrow \infty$$

In words:

As n (sample size) becomes large, the distribution of the sample mean becomes approximately standard normal, i.e., $N(0,1)$.

Example:

An electrical firm manufactures light bulbs that have a length of life that is approximately normally distributed, with a mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a random sample of 16 bulbs will have an average life of less than 775 hours.

$$\begin{aligned} Z &= \frac{\bar{X} - 800}{\frac{40}{\sqrt{16}}} \\ &= P(\bar{X} < 775) \\ &= P(Z < -2.5) \\ &= 0.0062 \end{aligned}$$

Example:

Given the population 1,1,1,3,4,5,6,6,6 and 7. Find the probability that a random sample of size 36 selected with replacement will yield a sample mean between 3.26 and 4.74.

$$Z = \frac{\bar{X} - 4}{0.373}$$

$$P(3.24 < \bar{X} < 4.74)$$

$$= P(-1.98 < Z < 1.98)$$

Example:

Given the population 1,1,1,3,4,5,6,6,6 and 7. Find the mean and standard deviation for the sampling distribution of means for a sample of size 4 selected at random with replacement.

Between what two values would you expect at least $\frac{3}{4}$ of the sample means to fall?

Example:

Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute.

Solution : In this case, $\mu = 28$ and $\sigma = 5$. We need to calculate the probability $P(\bar{X} > 30)$ with $n = 40$. Since the time is measured on a continuous scale to the nearest minute, an $\bar{x}$ greater than 30 is equivalent to $\bar{x} \geq 30.5$. Hence,

$$\begin{aligned} P(\bar{X} > 30) &= P\left(\frac{\bar{X} - 28}{5/\sqrt{40}} \geq \frac{30.5 - 28}{5/\sqrt{40}}\right) = P(Z \geq 3.16) \\ &= 0.0008. \end{aligned}$$

Estimation

Introduction

The process of drawing inferences about a population on the basis of information contained in a sample taken from the population is called **Statistical Inference**.

The statistical inference is divided into two areas:

Estimation of Parameters: Estimate the unknown value of a population by using the sample observations from the population.

Testing of Hypothesis: Enables us to decide on the basis of information obtained by sampling whether to accept or reject any hypothesis regarding the value of the parameter.

Estimate and Estimator:

An Estimator is the rule or method that is used to estimate a parameter.

Whereas, an estimator is the numerical value obtained by substituting the sample observations in the rule or the formula.

Estimation by Confidence Interval

The confidence interval estimate of an unknown parameter θ is an interval computed from a random sample of n values $x_1, x_2, \dots, x_n$ that contains the unknown parameter θ .

With α as a confidence coefficient, the interval is measured as $100(1 - \alpha)\%$ confidence interval.

Confidence Interval Estimate of a Population Mean of Normal Population with σ known

For a normal distribution of measurements with unknown mean μ and known variance σ^2 , a $100(1 - \alpha)\%$ confidence interval of μ follows from the probability

$$P \left[-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2} \right] = 1 - \alpha$$

Or,

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right and α is called the **confidence coefficient** or **confidence level**.

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where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right and α is called the **confidence coefficient** or **confidence level**.

Some frequently used values of $z_{\alpha/2}$ with respect to the percent confidence interval we want to calculate:

1.96 for 95%, 1.645 for 90%, 2.576 for 99%

Example:

The standard deviation of the amount poured into bottles by an automatic filling machine is 1.8 ml. The amounts of fill in a random sample of bottles, in ml, were 480, 479, 482, 480, 477, 478, 481, and 482. Suppose the population of the amount of fill is normal. Construct a 90% confidence interval for the mean amount in all bottles filled by the machine.

The confidence interval for the mean is

$$\bar{x} \pm \frac{Z_{\alpha/2} \sigma}{\sqrt{n}}$$

$$\bar{x} = 480, \sigma = 1.8, n = 8, z_{0.05} = 1.645$$

Hence, (478.95, 481.05).

Example:

A confidence interval is constructed from a sample of size 25, for the mean of normal population which has $\sigma = 50$. The limits for the interval are 110.2 and 135.8. What confidence coefficient was used?

$$\begin{aligned}\bar{x} - z_{\alpha/2} \frac{50}{5} &= 110.2 \quad \text{and} \quad \bar{x} + z_{\alpha/2} \frac{50}{5} = 135.8 \\ \Rightarrow z_{\alpha/2} &= 1.28 \\ 1 - \alpha &= 0.7994\end{aligned}$$

Hence, an 80% confidence was used.

Confidence Interval Estimate of a Population Mean of Normal Population with σ unknown

For a normal distribution of measurements with unknown mean μ and unknown variance σ^2 , we use the sample variance s^2 in place of σ^2 .
100(1 - α)% confidence interval of μ follows from the probability

$$P \left[-z_{\alpha/2} < \frac{\bar{X} - \mu}{s/\sqrt{n}} < z_{\alpha/2} \right] = 1 - \alpha$$

Or,

$$\bar{x} - z_{\alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{s}{\sqrt{n}},$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right and α is called the **confidence coefficient** or **confidence level**.

Example:

The Punjab Highway Department is studying traffic pattern on G.T. Road near Lahore. As part of the study, the department needs to estimate the average number of vehicles that pass the Ravi bridge each day. A random sample of 64 days gives $\bar{x} = 5410$ and $s = 680$. Find the 90 percent confidence interval estimate for μ , the average number of vehicles per day.

Confidence Interval Estimate of a Population Mean of Normal Population with σ unknown

If $X_1, X_2, \dots, X_n$ are normally distributed random variables with μ mean and σ^2 variance, then a $(1 - \alpha)100\%$ confidence interval for the population mean for unknown σ is:

$$\bar{x} \pm t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right)$$

This interval is often referred to as the "**t-interval for the mean.**"

Example:

A random sample of 16 Americans yielded the following data on the number of pounds of beef consumed per year:

118 115 125 110 112 130 117 112 115 120 113 118 119 122 123 126

What is the average number of pounds of beef consumed each year per person in the United States?

Solution: $\bar{x} = 118.44, s = 5.66, n = 16, t_{0.025,15} = 2.1314$

The interval is

(115.42, 121.46)

Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions or inferences about a population based on sample data.

It helps answer questions such as:

- *“Is this model’s accuracy really better than 85%?”*
- *“Do two datasets have the same mean?”*
- *“Is this variable significantly related to the output?”*

Hypothesis Testing

Hypothesis testing

We start with two competing statements about a population parameter (like the mean, variance, or proportion)

Symbol	Name	Meaning
H_0	Null hypothesis	The default assumption (no effect, no difference).
H_1 or H_a	Alternative hypothesis	What we want to test or prove (there <i>is</i> an effect).

Example

Suppose we train an AI model and want to test if its mean accuracy is more than 90%.

$$H_0: \mu = 0.90 \text{ (no improvement)}$$

$$H_1: \mu > 0.90 \text{ (model improves)}$$

Example (Continued)

A model claims to have 90% accuracy. You test it on 100 samples and find accuracy $\bar{x} = 0.94$, with known $\sigma = 0.1$.

We test:

$$H_0: \mu = 0.90, H_1: \mu > 0.90$$

Significance level: $\alpha = 0.05$

Test statistic (Z-test):

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{0.94 - 0.90}{0.1/\sqrt{100}} = \frac{0.04}{0.01} = 4$$

Decision:

Critical value at $\alpha=0.05$ for one-tailed test: $z_{0.05} = 1.645$

Since $4 > 1.645$, reject H_0 .

Conclusion:

There is strong evidence that the model's accuracy **is greater than 90%**.

Example:

A machine-learning model used to take **10 hours** on average to train.

After optimization, a data scientist claims the **average training time has changed** (could be shorter *or* longer).

You run **a sample of 25 training runs** and find:

$$\bar{X} = 9.4 \text{ hours}, s = 1.2 \text{ hours}$$

We will test this claim at **$\alpha = 0.05$** significance level.

Solution:

$H_0: \mu = 10$ (the mean training time has not changed)

$H_1: \mu \neq 10$ (the mean training time has changed)

This is a **two-tailed test** because we're checking for any change — higher or lower.

Determine the test statistic

Population standard deviation σ is **unknown**, so we use a **t-test**:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

Substitute values:

$$t = \frac{9.4 - 10}{1.2/\sqrt{25}} = \frac{-0.6}{0.24} = -2.5$$

Find critical t-values

For $\alpha = 0.05$ and $df = n - 1 = 24$,
the two-tailed critical values are:

$$t_{0.025,24} = \pm 2.064$$

Decision rule

If $t < -2.064$ or $t > +2.064$, **reject** H_0 . Otherwise, **fail to reject** H_0 .

Here $t = -2.5$, which is **less than -2.064**, so we **reject** H_0 .

Example:

Boys of a certain age are known to have a mean weight of $\mu = 85$ pounds. A complaint is made that the boys living in a municipal children's home are underfed. As one bit of evidence, $n = 25$ boys (of the same age) are weighed and found to have a mean weight of $\bar{x} = 80.94$ pounds. It is known that the population standard deviation σ is 11.6 pounds (the unrealistic part of this example!). Based on the available data, what should be concluded concerning the complaint?